MIL-A-45940(WC)

7 December 71

SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

ARMAMENT SUBSYSTEM, HELICOPTER, 7.62MM
MACHINE GUN: DOOR MOUNTED, LIGHT WEIGHT, M24
(Less M60D Machine Gun)

1. SCOPE

1.1 This specification covers one type light weight armament subsystem providing two pintle mounted 7.62mm, M60D machineguns for right hand door and left hand escape hatch installation on CH-47 type helicopters.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military		
MIL-P-116	-	Preservation, Methods of.
MIL-I-6866	-	Inspection, Penetrant Method of.
MIL-I-6868	-	Inspection Process, Magnetic Particle.
MIL-W-8611	-	Welding, Metal Arc and Gas, Steel and Corrosion and Heat Resistant Alloys; Process for.
MIL-W-12332	-	Welding, Resistance, Spot, Seam and Projection; For Fabricating Assemblies of Low-Carbon Steel.
MIL-W-13855	-	Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of.

FSC 1005

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STANDARDS

Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-1261	-	Welding Procedures and Constructional Steels.

DRAWINGS

U.S. Army Weapons Command		
A11699676	-	Armament Subsystem, Helicopter, 7.62mm Machine Gun, Door Mounted, Light Weight, M24.
IEL11013168	-	Index of Inspection Equipment Lists.

(Product drawings referenced in this specification are specified on Drawing A11699676 and related assembly drawings; acceptance inspection equipment drawings form a part of Drawing IEL11013168.)

PUBLICATIONS

U.S. Army Weapons Command		
P8427091	-	Packaging Data Sheet for Armament Subsystem, Helicopter: M24.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirement for submission of the first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The armament subsystem and parts shall conform to the materials, construction and design requirements specified herein, on Drawing A11699676 and drawings applicable thereto, and shall be in accordance with the applicable materials, construction and design provisions of MIL-W-13855.

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3.2.1 Welding.

3.2.1.1 Certification of welding. Unless otherwise specified (see 6.1), prior to production of any weldments, the contractor or manufacturer shall have Government approval of welding procedures and test specimens. Welding of parts and subassemblies shall be in accordance with the applicable drawing and as specified herein.

3.2.1.2 Welded joints. All welded joints of parts and subassemblies shall be sound and free of cracks. Welded joints shall be magnetic particle inspected in accordance with MIL-I-6868 and penetrant inspected in accordance with MIL-I-6866 as applicable.

3.2.2 Gun mount assembly.

3.2.2.1 Gun mount. The right and left hand gun mounts shall be capable of assembly for the right and left side door mounting respectively and be securely attached to the door mount by the quick release pin. Welding of the gun mount assembly shall be in accordance with MIL-W-8611.

3.2.2.2 Pintle. The pintle shall assemble in the pintle post and be retained in the pintle post by ball bearings. The pintle shall rotate throughout its full range of travel without binding. The ball bearing retainer set screw shall be securely staked in place.

3.2.2.3 Quick release pin. The quick release pin shall readily assemble to and disassemble from the mount bracket, and when assembled shall be retained in the mount bracket by the ball detent. The quick release pin shall be attached to the mount bracket by the cable (see 3.2.7).

3.2.3 Ammunition box. The ammunition box shall readily assemble to and disassemble from the weapon and be retained on the weapon by the weapon bracket assembly. The ammunition retaining rod shall move freely throughout its full range of travel and held securely in position by the spring clip. The ejector plunger shall move freely throughout its full range of travel and be returned to its original position by spring action. Welding of the ammunition box shall be in accordance with MIL-STD-1261, and shall withstand the weld peel test in accordance with MIL-W-12332.

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3.2.4 Empty cartridge bag. The empty cartridge bag shall be capable of being retained and removed from the cartridge bag frame assembly.

3.2.5 Empty cartridge bag frame assembly. The empty cartridge bag frame assembly shall be capable of being attached to and removed from the weapon. The latch shall move freely through its full range of travel without binding, and when released shall return to its original position by spring action. Welding of the cartridge bag frame shall be in accordance with MIL-STD-1261.

3.2.6 Elastic cord assembly. The elastic cord assembly shall be securely retained on the mount bracket. The elastic cord shall withstand a gradual applied stretch load of 75 pounds ± 5 pounds without damage or permanent stretch.

3.2.7 Cable assembly. The crimped sleeves on the cable shall be capable of withstanding the tensile load of 50 pounds without evidence of the cable pulling out of the sleeve.

3.2.8 Endurance. The armament subsystem shall have a 90% probability of firing 10,000 rounds, and withstanding 100 cycles of rough handling without malfunctions, unserviceable parts or cracking of welded joints with a 10% confidence level. Testing shall be as specified in 4.4.3.2 and 4.6.1. A malfunction is defined as one which a failure of the mount as a component of the mount precludes further firing without replacement of the mount or component of the mount.

3.2.9 Targeting. The armament subsystem with M60D machinegun assembled to both right hand mount and left hand mount shall meet the target requirements defined in Figures 1a, 1b, 2a and 2b (see 4.4.2.2.5).

3.3 Marking. Each armament subsystem shall be clearly marked in accordance with the applicable drawings and MIL-W-13855.

3.4 Workmanship. Workmanship shall be in accordance with the workmanship requirements of MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 First article inspection. First article shall be selected from early production and submitted for inspection in accordance with the contract requirements (see 6.1). The first article shall be representative of production processes to be used during quantity production. The first article shall be subjected to all examination and test specified herein and such other examination and test as are necessary to determine compliance with the requirements of the contract.

4.4 Quality conformance inspection.

4.4.1 Inspection lot. The formation, size and presentation of inspection lots of armament subsystems, parts and packaging shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable in consideration of quality history, manufacturing conditions, and contractor's delivery schedule, and within the limitations of MIL-W-13855. Armament subsystems shall be assembled from lots of component parts that have met all inspection requirements specified herein.

4.4.2 Examination.

4.4.2.1 Component parts and concurrent repair parts. Examination of component parts and concurrent repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1). The contractor's examination of these parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.4.2.2 Armament subsystem. Final examination of armament subsystems shall be performed after completion of all testing and just prior to preservation and packaging. Unless otherwise specified herein, each armament subsystem shall be examined as specified below. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating and to determine the general quality, completeness of manufacture, assembly, clarity and legibility of markings and workmanship. Armament subsystems failing to meet the requirements shall be rejected.

4.4.2.2.1 Gun mount assembly.

a. Manually examine the gun mount pintle to assure that it is retained in the bearing by the ball bearings and that it can be manually rotated throughout its full range of travel without binding.
b. Visually examine to assure the ball bearing retaining set screw is securely staked in place.

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c. Manually examine the quick release pin to assure that it readily assembles to and disassembles from the mount bracket and when assembled, is retained in the mount bracket by the ball detents.

d. Visually and manually examine to assure that the quick release pin is securely attached to the mount bracket by the cable (see F11699655 and F11699658).

e. Visually and manually examine the elastic cord assembly to assure that it is securely retained on the mount bracket and that it has the full range of swing (see F11699655 and F11699658).

f. Visually and manually examine all welded joints to assure secure or missing welds.

4.4.2.2.2 Ammunition boxes.

a. Visually and manually examine the retaining rod to assure that it moves freely throughout its full range of travel and that it is held securely by the spring clip.

b. Visually and manually examine the plunger to assure that it slides freely throughout its full range of travel and when released is returned to its original position by spring action.

c. Visually and manually examine all riveted joints to assure secure riveting and that there is no relative movement of the riveted parts.

d. Visually and manually examine all welded joints to assure secure or missing welds.

4.4.2.2.3 Empty cartridge bag.

a. Visually and manually examine the empty cartridge bag to assure that the latch moves freely throughout its full range of travel and when released is returned to its original position by spring action.

b. Visually and manually examine all welded joints of the cartridge bag frame assembly to assure secure or missing welds.

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4.4.2.2.4 Armament subsystem assembly.

a. Visually and manually examine all welded joints to assure secure or missing welds.
b. Examine assembleability of the machinegun to the gun mount pintle assembly using approved inspection equipment that simulates the maximum mounting conditions of the M60D machinegun.
c. Using a Government standard M60D machinegun, and equipment as depicted on Figures 1a, 1b, 2a and 2b, examine to assure that the elevating and traversing limits of the armament subsystem are within the requirements indicated on Figures 1a, 1b, 2a and 2b as applicable for right and left hand mount assemblies. Control of components shall be exercised to meet these limits. Failure to meet these limits shall be cause for rejection and the corrective action taken in accordance with 4.4.3.1.

4.4.2.3 Packaging. Examination of packaging of armament subsystems shall be performed in accordance with the classification of defects and acceptable quality levels (AQL's) specified in 4.4.2.3.1. Sample size shall be in accordance with MIL-STD-105, using inspection level I. The following provisions shall apply:

a. The AQL's are specified as percent defective.
b. The AQL listed for each defect shall be applied to the individual defect, not to a group of defects.
c. Examination for packaging defects specified in 4.4.2.3.1 shall apply to each item of the applicable sample of armament subsystems, interior packages, or exterior containers, as applicable.

4.4.2.3.1 Classification of defects for packaging. (Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet P8427091).

Categories	Defect	AQL
Critical:	None defined.	
Major:		
101	Illegible or incorrect marking.	1.0
102	Improper location of marking.	1.0
103	Improper level of packaging or packing (see procurement documents).	1.0
104	Inadequate cleaning and drying.	1.5
105	Improper preservation application.	1.5
106	Improper closure of unit package.	1.5
Minor:	None defined.	

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4.4.2.4 Armament subsystems which have passed all examinations after successful completion of testing shall be stamped with the Department of Defense complete inspection approval stamp on each of the base assemblies next to the pintle post assembly holes. The Government representative will control the stamps used for such stampings and observe the stamping operations as required to assure successful completion of all required examination and tests.

4.4.3 Testing.

4.4.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete armament subsystem inclusive of Government Furnished Property. If test requirements cited herein are not met, acceptance of the subsystem shall be deferred and the contractor shall accomplish as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c. Examine subsystems, partially assembled subsystems and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.

d. Upon instructions from USAWECOM, ATTN: AMSWE-QA, resume testing of the subsystem or retest an additional subsystem from the same lot.

4.4.3.2 Endurance testing.

4.4.3.2.1 Lot size. The endurance test lot shall consist of 50 armament subsystems.

4.4.3.2.2 Procedure. One mount assembly (one-half of the armament subsystem) selected by the Government representative from each endurance test lot shall be tested by the contractor using the test method specified in 4.6.1. Mount assemblies selected for the endurance test may be either left hand mounts or right hand mounts and shall be a representative of all armament subsystems produced. If the endurance requirements of 3.2.8

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are not met, acceptance of armament subsystems shall be deferred and the contractor shall perform the corrective actions as specified in 4.4.3.1. The Government representative shall verify the results of failure analysis and assure that corrective action, as applicable, has been accomplished. Malfunctions attributable to the machinegun, ammunition, or links shall not be counted against the armament subsystem being tested, however, such malfunctions shall be recorded.

4.4.3.3 Packaging testing. The contractor shall furnish the Government representative with certification that the packaging materials conform to the applicable packaging data sheets and specifications.

4.4.3.3.1 Determination of cleanliness testing. The contractor shall test items from each inspection lot for determination of cleanliness using the test method specified in 4.6.3.1. Sampling shall be in accordance with MIL-P-116.

4.4.3.3.2 Heat-sealed seam and vacuum retention testing. The contractor shall test level A unit packages from each inspection lot for heat-sealed seam and vacuum retention using the test methods specified in 4.6.2.2 and 4.6.2.3 respectively. Sampling shall be in accordance with MIL-P-116.

4.4.3.3.3 Repair parts. Testing of packaging of repair parts shall be performed in accordance with the criteria specified in the contract (see 6.1).

4.5 Inspection equipment. Unless otherwise specified in the procurement documents (see 6.1), responsibility for acquisition, maintenance and disposition of measuring and testing equipment prescribed on lists contained on the Index of Inspection Equipment Lists, Drawing IEL11013168 and for all other inspection equipment required to perform inspection required by applicable specifications, shall be in accordance with MIL-I-45607.

4.6 Test method.

4.6.1 Endurance test. Testing of armament subsystems for endurance shall be accomplished with the mount assembly attached to a fixture simulating the mounting structure of the helicopter. A Government standard M60D machinegun shall be used for the test (see 6.1). Ammunition used for the test shall be Government standard 7.62mm M59 or M80 ball cartridges linked with Government standard 7.62mm M13 links (see 6.1).

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4.6.1.1 A total of 10,000 rounds shall be fired from the initial test mount assembly regardless of the number of malfunctions or unserviceable parts requiring replacement in order to complete the test.

4.6.1.2 The 200 round rapid reload ammunition box shall be attached to the machine gun. The ammunition box shall be loaded with ammunition cartons, each containing 100 rounds linked ammunition. The two (2) 100 round belts shall be linked together with a Government standard 7.62mm dummy inert cartridge to form one (1) continuous belt. The empty cartridge bag shall be attached to the machinegun to catch and retain the fired cartridge cases and links.

4.6.1.3 Firing shall be accomplished in 200 round series. The first 100 rounds of each series shall be fired in interrupted bursts with at least 10 intentional breaks in firing. The second 100 rounds of each series shall be fired in one continuous burst.

4.6.1.4 In addition to the firing, the armament subsystem shall be tested for rough handling by placing the mounted machinegun, with a loaded ammunition box in the maximum position of depression. From this position, the machinegun shall be released and allowed to fall of its own weight throughout the full range of travel on the mount assembly. This shall be accomplished twice following each 200 round series.

4.6.1.5 The machinegun shall be cooled to ambient temperature after each 200 rounds of firing using cooling aids other than water. The machinegun may be cleaned and oiled after each 2,000 rounds of firing. At the close of each day's firing, the machinegun and mount assembly shall be protected against corrosion.

No mount assembly parts shall be altered and only parts broken or worn to the extent that they are unserviceable shall be replaced.

4.6.1.6 Welded joints of the mount assembly shall be checked for cracks using the magnetic particle inspection process in accordance with MIL-I-6868 and penetrant inspection process in accordance with MIL-I-6866 as applicable before and after the endurance test.

4.6.1.7 Complete accurate records shall be kept for each endurance test, showing each malfunction and part replacement including the number of the round at which each occurred and corrective action taken.

4.6.1.8 Disposition of the endurance tested mount assemblies shall be as specified in the contract (see 6.1).

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4.6 2 Packaging tests.

4.6.2.1 Determination of cleanliness. The surface of each sample unit shall be subjected to the determination of cleanliness test specified in MIL-P-116, except that the wipe test shall not be applicable to parts with black oxide or anodized protective coatings.

4.6.2.2 Heat-sealed seam. The sample level A unit packages shall be subjected to the heat-sealed seam test specified in MIL-P-116.

4.6.2.3 Vacuum retention. A sufficient vacuum shall be drawn to cause the flexible barrier of level A unit packages to cling snugly to the enclosed item. Care shall be exercised to insure that an excessive amount of vacuum is not applied which might cause puncture or rupture of the barrier.

4.6.2.3.1 Interpretation of results. After remaining undisturbed at room temperature for two hours, the barrier shall be examined to determine whether it is still taut and retracts against the item when drawn away and quickly released.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot pack consisting of complete and packed unit, packaged and packed in accordance with Packaging Data Sheet P8427091 to the level of protection specified in the contract (see 6.1) and forwarded as specified in 3.1.

5.2 Preservation, packaging, packing and marking. The armament subsystem with equipment shall be preserved, unit packaged, packed and marked in accordance with the requirements of Packaging Data Sheet P8427091 for the level of protection specified (see 6.1).

5.3 Repair parts. Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets specified in the contract (see 6.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Lists of drawings and specifications pertinent to the armament subsystem, showing applicable revision dates.
c. Inspection criteria for components (see 4.4.2.1).
d. Index to inspection equipment list pertinent to the armament subsystem showing applicable revision dates.

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e. Number of first article samples required (see 3.1).
f. That packages opened for examination shall be repackaged by the contractor at the contractor's expense.
g. Disposition of endurance tested mounts (see 4.5.9.6).
h. Shipping instructions for first article samples and pilot pack, if different (see 3.1 and 5.1).
i. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
j. Packaging instructions for repair parts (see 5.3).
k. Place of final inspection and acceptance (see "Notes" in MIL-W-13855).
l. List of acceptance inspection equipment to be furnished the contractor and responsibilities for other Government property to be furnished the contractor (see 4.5 and 6.8).
m. Responsibilities for furnishing ammunition, links and machinegun (see 4.6.1).
n. Certification of welding procedures and test specimens if different (see 3.2.1).
o. Procedures and methods for demilitarizing and disposing of rejected material.
p. Disposition of Government furnished property.
q. Responsibility for test firing facilities and operating procedures (see 6.2).

6.2 Test firing facilities and operating procedures shall be designed by the contractor in conformance with local, state, and federal regulations and suitable for carrying out prescribed firing tests with safety of operating and visiting personnel. Copies of these contractor designs shall be forwarded to the contracting officer. Government facilities may be viewed upon application to the contracting officer.

6.3 When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

6.4 To avoid delay in test firing, the contractor should maintain a minimum of two months' supply of ammunition as determined by anticipated firing requirements.

6.5 Contract data requirements. Monthly reports of the results of final examination and endurance test shall be specified for delivery on the DD Form 1423 included in the contract.

6.6 Supersession data. This specification includes the requirements of Springfield Armory Purchase Description SAPD-280A, dated 6 August 1965.

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6.7 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.8 Unless otherwise specified (see 6.1 1), the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Summary List, DD Form 1660.

Custodians:
Army - WC

Preparing activity:
Army - WC

Project number:
1005-A422

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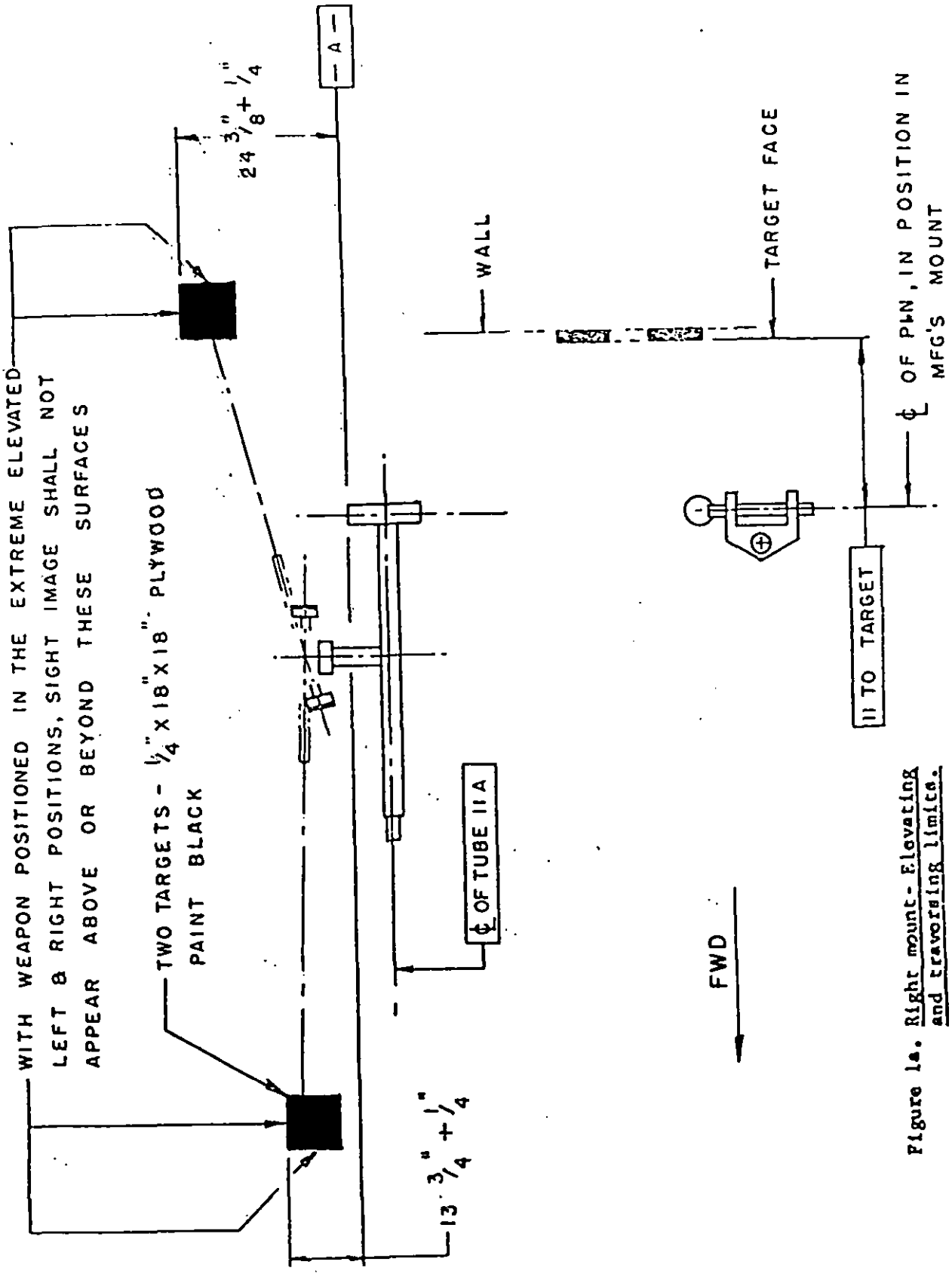


Figure 1a. Right mount- Elevating and traversing limits.

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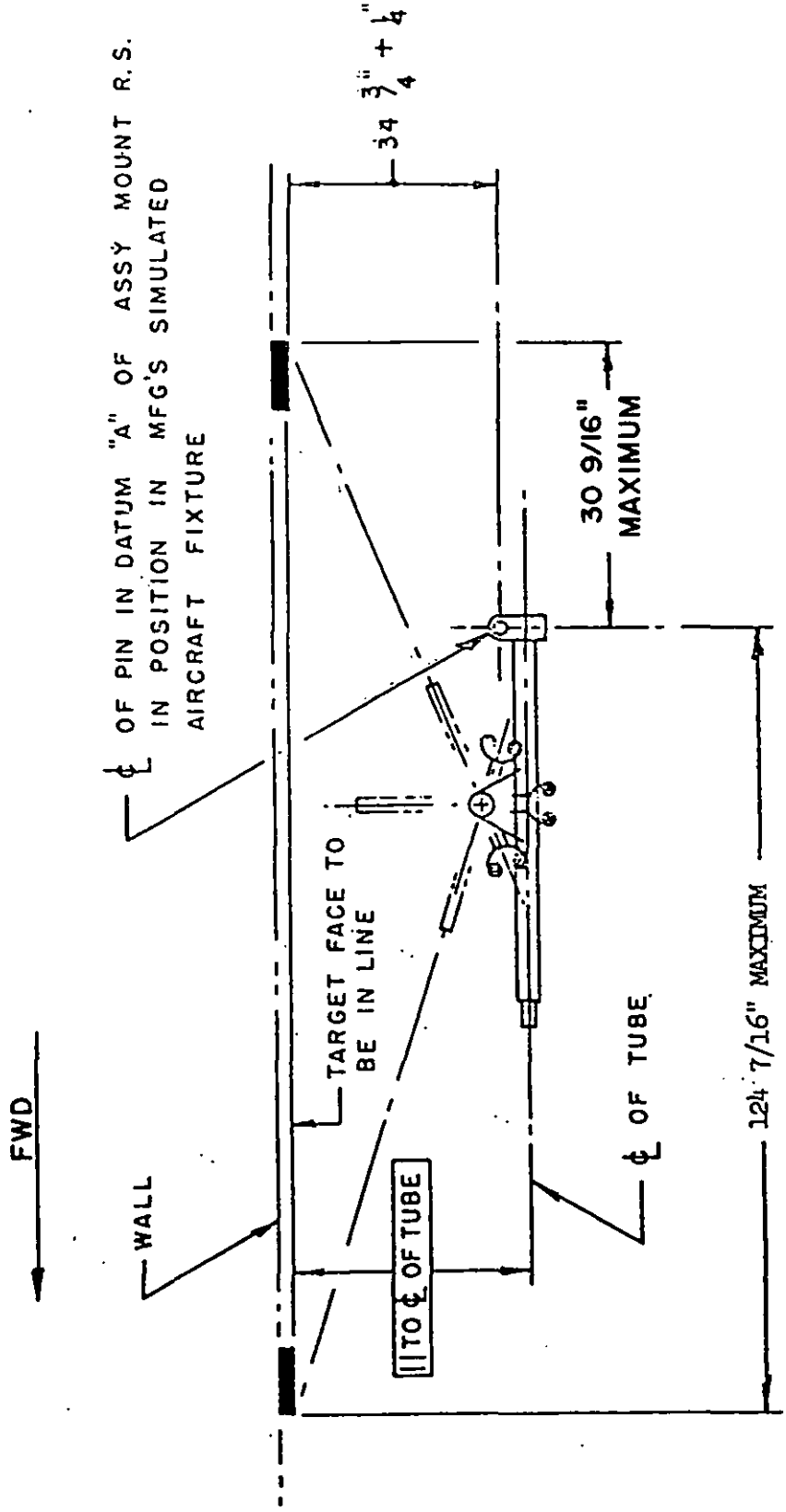


Figure 1b. Right mount - Elevating and traversing limits.

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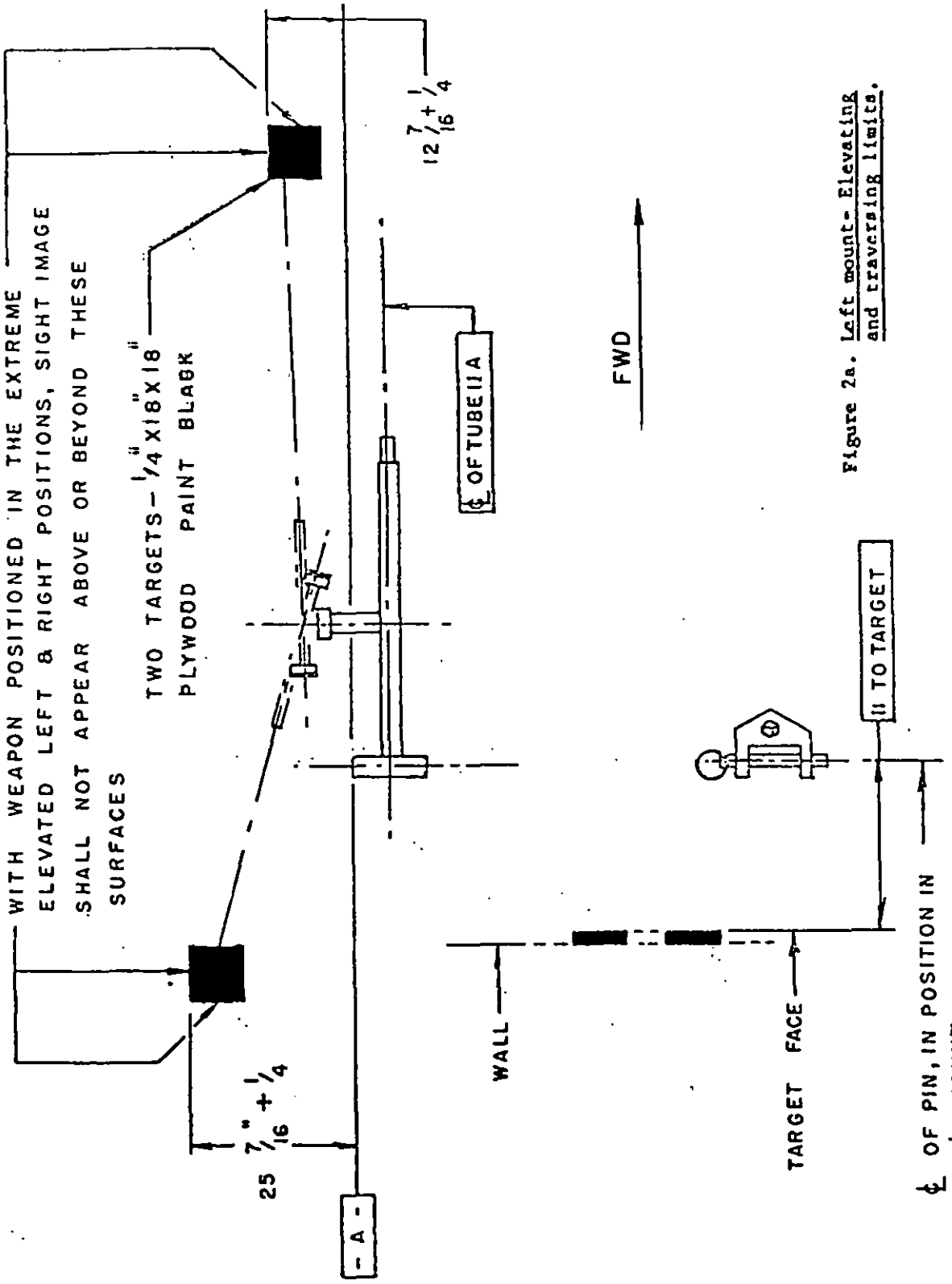


Figure 2a. Left mount- Elevating and traversing limits.

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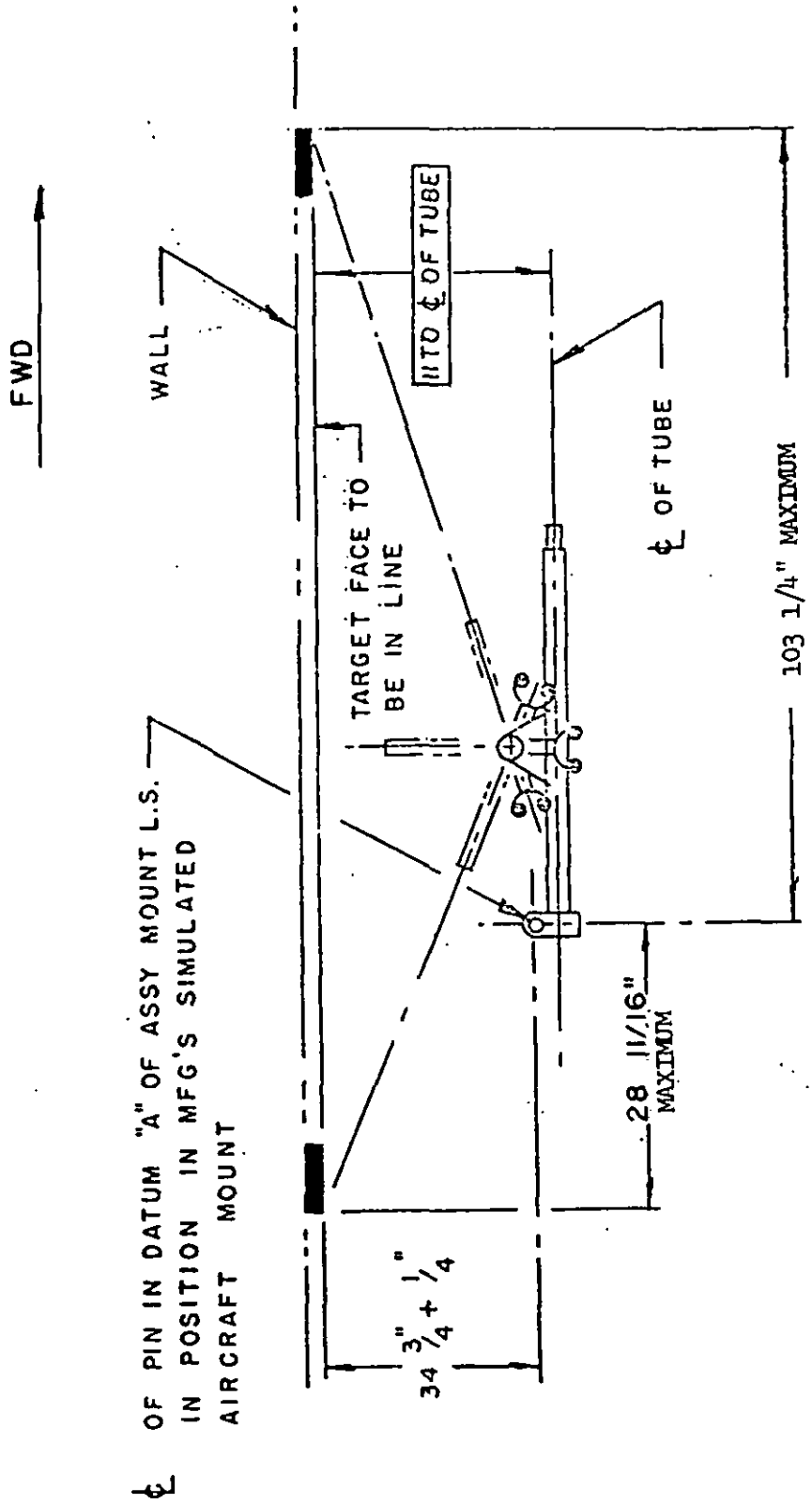


FIGURE 2b. Left mount - Elevating and traversing limits.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. *Comments and the return of this form* will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions *submitted on this form do not constitute or imply* authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL *PROCURED UNDER A*

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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